

Quiz 2

Each question is worth 3 to 4 points. Your score is total of best 5.

1. Find $|GL_3(\mathbb{F}_5)|$, i.e., the number of invertible 3×3 matrices with entries in $\mathbb{F}_5$, the field consisting of integers mod 5. You may leave your answer as a product of positive integers (each expressed in ordinary base 10 representation), but be sure to explain your reasoning.

2. Suppose A and B are subspaces of a vector space V . The dimensions of A , B and V are 22, 10, and 25 respectively.

- (a) Is it possible for the sum $A + B$ to be direct? If yes, give an example. If not, state why.
- (b) Let $A \oplus B$ be the external direct sum of A and B , i.e., $A \oplus B =$ the set $A \times B$ with addition and scalar multiplication defined slotwise. The map $f : A \oplus B \rightarrow V$ defined by $f(a, b) = a + b$ is linear. (Check this in your mind if you wish. It is not part of the question.) What is the kernel of this map?
- (c) Make a table with two columns as follows. In the first column list all possible values of the dimension of the subspace $A + B$ of V . For each entry in the first column, write on the same line in the second column all possible corresponding values of the dimension of $A \cap B$. Explain.

3. Let $M = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 1 & 3 \end{bmatrix}$. Solve this problem without using any polynomial or variables.

- (a) Find a basis of each of the following starting with the definition. 1-eigenspace of M , 3-eigenspace of M , 5-eigenspace of M .
- (b) Either show why M cannot be diagonalized OR diagonalize M explicitly, i.e., write a suitable equation involving M , a diagonal matrix D and possibly one or more additional matrices, where all matrices must have explicit constants as entries (no unknowns).

4. Solve the following independent questions.

- (a) Find the characteristic polynomial of the matrix $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.
- (b) The characteristic polynomial of a 4×4 matrix M factors fully and has two distinct roots. Write one such matrix M for which the sum of dimensions of its eigenspaces has the minimum possible value. Explain briefly.

5. An operator T on a vector space of dimension 100 satisfies the following. For each positive integer $k \leq 100$ the dimension of the kernel of T^k is k . What is the rank of T ? Show how we can get a basis of V with respect to which the matrix of T has the minimum possible number of nonzero entries. Justify.

6. Let $p(x) = (x - 22)(x - 10)(x - 25)$. Suppose an operator S on a vector space W satisfies $p(x)$ but satisfies no proper factor of $p(x)$. Show that each of 22, 10, 25 must be an eigenvalue of S . (To think about afterwards: must S be diagonalizable?)